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IISE PG&E Energy Analytics Challenge 2024: Forecasting day-ahead electricity prices

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ABSTRACT

Electricity price forecasting is one of the cornerstones of modern-day power system operation. Over the last two decades, the field has seen many methodological advancements. On one end, traditional statistical methods remain widely prevalent in today's energy industry, as they have proven consistently to “stand the test of time.” On the other end, emerging predictive techniques (e.g., machine and deep learning) have demonstrated immense potential. Yet, due to limited benchmarking studies, the magnitude of improvement realized by such emerging methods relative to their statistical counterparts is not entirely clear. In response, two technical divisions of the Institute of Industrial & Systems Engineers (IISE) partnered with the Pacific Gas and Electric Company (PG&E)—one of the largest utility companies in the United States—in order to organize an electricity price forecasting challenge in 2024. Using three years of pricing signals and exogenous information from California's electricity market, the competition challenged teams of researchers and practitioners to design their own models and submit forecasts for day-ahead electricity prices, which were independently evaluated against a test set that has been withheld from all teams. This article introduces the challenge, as well as an overview of the methods used by the top-performing contestants. A distilled summary of the key insights and lessons learned by the challenge organizers is then presented, together with their relevance to the topic of electricity price forecasting. This is then followed by recommendations for future similarly focused competitions. To accelerate the research and development on this important topic, all data used in the challenge have been made publicly available by the challenge organizers.

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1. Introduction

Forecasting electricity prices is a routine, yet highly important task in modern-day power systems. In one way or another, almost every stakeholder in the power and energy industry relies on electricity price information and forecasts to inform their business decisions. **As a case in point, day-ahead forecasts of electricity prices are used on a daily basis by energy traders in de-regulated electricity markets to determine their bidding strategies and maximize the profitability of their businesses (Li *et al.*, 2007; Chen *et al.*, 2023).** **Another example is the use of long-term electricity price forecasts by power producers to adjust their assets' production, maintenance, and outage schedules to minimize revenue losses (Yildirim *et al.*, 2016; Papadopoulos *et al.*, 2023).** The value of accurate electricity price information is further substantiated by the rapid diversification of today's electricity system in terms of its constituting energy sources, giving rise to increased volatilities and uncertainties that make the management of the electricity system more complicated than ever.

The operation of power systems, in light of the increasing penetration of Variable Renewable Energy (VRE) resources,

requires characterizing multi-source uncertainties. Three major sources of uncertainty are, in no particular order of importance, the electric load, supply, and pricing. Perhaps the most classic among the trio, electric load is a long-standing random quantity in electricity system operation. The uncertainty in electric load stems from the variability in the demand of electricity across time (e.g., seasons, months, and days), and space (i.e., geographical locations). An extensive body of literature has been dedicated to modeling the variations in electric loads, including in the context of emerging operational decisions such as demand response management and peak load shaving (Hong and Fan, 2016; Hong and Wang, 2022). The second source of uncertainty is electric supply. Modeling and forecasting electric supply is becoming more and more challenging, especially as weather-dependent VRE resources continue to be integrated into the grid at increasing levels (Ding, 2019; Ye *et al.*, 2024a). This includes but is not limited to, models and methods to forecast wind and solar power generation at a multitude of spatial and temporal scales (Ezzat *et al.*, 2018; Zhang *et al.*, 2024). The final element of the “uncertainty trio” is electricity pricing, which defines the value of a unit of electricity at a given

time and location. Several external factors can affect electricity pricing, including, but not limited to, local weather patterns, grid condition, the variations in the electricity generation mix, economic and political drivers, as well as the time of day.

Understandably, the three sources of uncertainty, namely electric load, supply, and pricing, are by no means independent. For example, the pricing of electricity is largely governed by the interplay between the electric load and supply across space and time (Pinheiro *et al.*, 2023). This is especially true considering that electricity, unlike other commodities, is not yet economically storable at a large scale (Deguenon *et al.*, 2023). The focus of this article is on electricity price forecasting for short-term look-ahead horizons (a few hours up to a day ahead).

The modern literature on short-term electricity price forecasting started to formally consolidate in the early 2000s as a response to the deregulation of electricity markets in the late 1990s (Weron, 2014). Today, existing methods for short-term electricity price forecasting can be broadly grouped into two categories. The first category is primarily rooted in classic statistical methods. Those include, but are not limited to, dynamic regression and autoregressive-based approaches (Conejo *et al.*, 2005; Zareipour *et al.*, 2006), multivariate and regularized linear regression (Uniejewski *et al.*, 2016; Ziel and Weron, 2018; Sgarlato and Ziel, 2023), exponential smoothing methods (Cruz *et al.*, 2011), and those based on regime-switching methods (Karakatsani and Bunn, 2008). The second category bears on the recent advancements in Machine Learning (ML) and Artificial Intelligence (AI). Relevant examples are tree- and ensemble-based approaches (Mei *et al.*, 2014; Wang *et al.*, 2022), as well as kernel-based methods (Chaâbane, 2014). Recently, several Deep Neural Network (DNN) architectures have been tailored for electricity price forecasting, including, for example, those based on recurrent (Xie *et al.*, 2018), convolutional (Lago *et al.*, 2018), and transformer (Zhang and Wu, 2023) networks. Weron (2014) lists additional classes of short-term electricity price forecasting models other than the statistical and ML/AI paradigms reviewed above including fundamental, structural, and multi-agent models.

Despite the progress in ML/AI methods (second category above), classic statistical approaches are still widely prevalent in the practice of short-term electric price forecasting. Rooted in rigorous regression and statistical forecasting theory, those methods have consistently stood the test of time across diverse case studies, extensive experimental setups, and numerous datasets. This makes them a fairly “safe” choice by domain experts and energy practitioners, relative to the emerging AI/ML techniques. Add to that the parsimonious nature and interpretability of those models, which significantly facilitate their adoption in practice using standard statistical software at minimal cost and background knowledge.

On the other hand, there is a large number of studies that report significant accuracy gains in forecasting electricity prices through novel AI/ML techniques. Yet, most of those studies reported those gains based on private or

difficult-to-access datasets that are not always available to the research community. The same issue was noted by other researchers in the electricity price forecasting literature (Lago *et al.*, 2021; Sheybanivaziri *et al.*, 2024). This raises a timely question: *what is the real potential or value of emerging ML/AI techniques in electricity price forecasting relative to traditional statistical methods?* An effective approach to answer this question is through a well-executed competition that *blindly* evaluates numerous methods on a test set that has been reserved from all contestants. This provides a much *fairer* evaluation than typical experimental setups in most scientific papers where access to the test data is available to the researchers as part of the original experimental design. Towards that, the Energy Systems (ES) and Quality Control and Reliability Engineering (QCRE) divisions of the Institute of Industrial & Systems Engineers (IISE) partnered with the Pacific Gas & Electric company (PG&E) to introduce the 2024 Energy Analytics challenge with a focus on day-ahead electricity price forecasting. Using three years of historical pricing signals and exogenous information from California’s electricity market, the objective of the competition was to challenge participants to design, test, and submit their models and electricity price forecasts of a representative 2-week period in a given year. The forecasts were then evaluated on a test set which had been reserved from all challenge participants. The submissions were primarily evaluated based on forecast accuracy, but also through other criteria that reward model simplicity, reproducibility, and innovation.

Data challenges and competitions have had a key role in advancing the science and practice of forecasting, even beyond the topic of electricity price prediction. One of the most famous examples are the Makridakis (M) Competitions organized since 1982 (Makridakis *et al.*, 2022a). In its sixth and most recent offering, the M6 competition focused on multivariate financial time series forecasting. In energy applications, the GEFCom 2017 (Hong *et al.*, 2016) and the BigDEAL 2022 (Shukla and Hong, 2024) competitions focused on electric load forecasting, whereas GEFCom 2012 and the Hybrid Energy Forecasting and Trading competitions focused on electric power forecasting. Table 1 lists a representative set of energy-focused challenges and their key features. The main emphasis of these competitions is to forecast one or multiple elements of the uncertainty trio (namely, load, supply, and pricing). The competitions typically select a quantitative metric to evaluate the quality of the forecasts (e.g., mean absolute error, continuous ranked probability score). Some competitions also consider model complexity and innovation as additional evaluation criteria. Other energy-focused competitions not listed in Table 1 include those where the focus shifts from forecasting to decision-making. For example, ARPA-E hosted a series of power system optimization challenges for solving large scale instances of the security constrained optimal power flow problem (Aravena *et al.*, 2023; ARPA-E, 2023; Oh *et al.*, 2023). Other operations-engineering-focused competitions have addressed “prescriptive” aspects in renewable energy generation such as storage scheduling under

Table 1. Summary of select energy-focused forecasting competitions

Competition	Forecast Variable	Target Frequency	Data Provided	Exogenous Data Allowed?	Evaluation Criteria	Year
IISE PG&E Energy Analytics Challenge	Price	Hourly	Load, Gas Price, Electricity Price	Yes	MAE, Model Design	2024
Hybrid Energy Forecasting & Trading (IEEE PES, 2023)	Renewable Generation	30 mins	Weather, Market Data, Power Generation	Yes	Pinball Loss	2023
Net Load Forecasting Prize (DoE, 2023)	Load	Hourly	Load	Yes	CRPS	2023
Hessian Energy Forecasting (Hessian, 2022)	Residual Load	15 mins	Weather, Load, Generation	No	RMSE, Model Design	2022
Demand Forecasting - Post-COVID (IEEE, 2020)	Load	Hourly	Demand,			
Weather	No	MAE				
GEFCOM 2017 (Hong <i>et al.</i> , 2019)	Load	Hourly	2020 Load	Yes	Pinball Loss	2017
GEFCOM 2014 (Barta <i>et al.</i> , 2015)	Load, Price, Generation	Hour	Load, Weather, Price	No	Pinball Loss	2014
GEFCOM 2012 (IEEE, 2012)	Load, Wind Power	Hour	Temperature, Load	No	MSE	2012

renewable uncertainty (NationalGrid, 2021), wind farm layout modeling (WFLOC, 2014), wind turbine performance prediction (WeDoWind, 2023), as well as operations and outage scheduling in energy networks (ROADEF, 2010).

Except for GEFCOM 2014 (Barta *et al.*, 2015) which had one of the competition tracks focused on electricity prices, the overwhelming majority of energy-focused competitions, as shown in Table 1, primarily focus on electric load and supply prediction. To complement and build on those competitions, the IISE PG&E 2024 Energy Analytics challenge introduced herein addresses the third element of the uncertainty trio, which is electricity price forecasting. We note, however, that despite our specific focus on modeling electricity prices, we envision several of our findings and insights to be transferable to other relevant forecasting problems in the energy industry, including both supply and load forecasting.

The remainder of this article is organized as follows. In Section 2, we introduce the IISE PG&E Energy Analytics Challenge, including the forecasting problem set up, data used, and evaluation criteria. Section 3 overviews the methods proposed by the top-performing teams, as well as how they fared against each other, and against a representative set of standard forecasting benchmarks that is widely used in the energy industry. Section 4 presents a discussion about the key insights and lessons learned by the challenge organizers and their relevance to the future of electricity price forecasting. The article is concluded in Section 5.

2. IISE PG&E Energy Analytics Data Challenge 2024

This section starts with introducing the aim of the competition, followed by a description of the dataset provided by PG&E, the forecasting problem setup, benchmarks, and evaluations.

2.1. Challenge aim and description

The aim of the challenge is to engage researchers and practitioners in designing innovative solutions for short-term electricity price forecasting. The specific focus is on modeling

and predicting day-ahead electricity prices in the California power grid operated by the California Independent System Operator (CAISO). The CAISO market comprises two settlement designs: (i) day-ahead (up to 7 days in the future), providing financial clearing and reliability planning considering net load forecasts, and (ii) real-time (sub-hourly), dispatching generation resources in near real-time. Of interest to this challenge is the day-ahead settlement design, wherein electricity market participants bid their assets, whether for buying electricity or selling generated power. In this challenge, we focus on 24-hour ahead price forecasts. For energy traders, an accurate forecast of electricity prices is key to ensure profitable market participation. For system operators, these forecasts enable proactive planning of operations to ensure the electric supply and demand are matched at minimal cost and satisfactory reliability levels.

The challenge specifically focuses on predicting the day-ahead electricity prices at the pricing node of NP-15, shown in Figure 1. Located in Northern California, NP-15 serves as a crucial geographic location for pricing electricity within the CAISO market. There are two distinct NP15 prices: one generated by the CAISO market processes and the other through a bilateral contract for differences against a series of CAISO market prices. The CAISO NP15 price is a weighted average of Locational Marginal Prices (LMPs) at supply nodes and does not directly influence the prices at individual supply nodes. The bilateral price acts as a hedge, where one counterparty agrees to pay a fixed price in exchange for the future realized average NP15 price set in the CAISO markets. Electricity prices often exhibit significant spatial variability across various pricing nodes, due to factors such as local supply and demand conditions, weather conditions, transmission constraints, and the generation mix. Prices in the day-ahead market are significantly influenced by supply and demand forecasts and the bidding in that market. These prices reflect the overall cost of electricity generation, including fuel costs, renewable resource availability, and transmission constraints. Contestants were expected to provide day-ahead hourly price forecasts. The model is assumed to be executed every day at 8:00 AM local time to predict the next day's hourly prices in light of the most recent available data.



Figure 1. Location of NP15 in Northern California, United States. ZP26 and SP15 are other pricing nodes within the CAISO market. Figure adapted from the official CAISO's Open Access Same-Time Information System (OASIS) as of December 2024 (CAISO, 2024).

Table 2. Variables included in the training dataset supplied to all challenge participants

Variable Name	Description
OPR_DATE	Record of the day, month, and year
HOUR_ENDING	Record of the hour within the day
LOADING_MW_ACTUAL_CAIISO	Electric load in CAISO (MW)
LOADING_MW_ACTUAL_PGE	Electric load in PG&E (MW)
LOADING_MW_ACTUAL_SCE	Electric load in SCE (MW)
LOADING_MW_ACTUAL_SDGE	Electric load in SDGE (MW)
LOADING_MW_FORECAST_CAIISO	Electric load forecast in CAISO (MW)
LOADING_MW_FORECAST_PGE	Electric load forecast in PG&E (MW)
LOADING_MW_FORECAST_SCE	Electric load forecast in PG&E (MW)
LOADING_MW_FORECAST_SDGE	Electric load forecast in SDGE (MW)
GAS_PRICE_PGE	Gas prices at PG&E (\$/unit)
GAS_PRICE_SCE	Gas prices at SCE (\$/unit)
DA_LMP_PGE_NP15	Day-ahead LMP at NP15 (\$/MW)

2.2. Data description

The dataset was supplied by PG&E and included 3 years of hourly day-ahead electricity prices from 2020 to 2022. In addition to pricing signals, PG&E provided information on electric loads and gas prices in CAISO, PG&E, Southern California Edison (SCE), and San Diego Gas & Electric (SDGE). Table 2 lists all variables in the training dataset, along with their description and units. Besides calendar and time information, there is a total of 10 exogenous variables (or features) representing electric loads and gas prices at different geographical locations. The target forecast variable is called DA_LMP_PGE_NP15, which denotes the day-ahead LMP at NP15.

Figure 2(a) shows the time series of the electric loads in CAISO, PG&E, SCE, and SDGE. The electric loads show clear diurnal and seasonal cycles with strong spatial correlations. Figure 2(b) shows the gas prices at PG&E and SCE.

Except for few large spikes, gas prices show small variations over time. The variability of gas prices at SCE appears to be higher than that at PG&E. Figure 2(c) show the electricity prices at NP15 (target variable). Unlike the electric loads, the electricity prices shows less clear temporal patterns, with several sporadic, short-duration spikes. Those spikes are often associated with extreme weather patterns and supply scarcity periods, and represent a major challenge to the contestants in designing their models. It is also noted that some of the electric price spikes appear to align with gas price spikes, suggesting coupling dynamics between electricity and gas networks (Byeon and Van Hentenryck, 2019). It is also worth mentioning that the last few weeks of the dataset show a noticeable increasing trend in the electricity and gas prices relative to the historical data (excluding sporadic spikes). This represents yet another data science challenge to the contestants in determining how much weight should be assigned to recent versus distant observations when making future short-term predictions. Overall, electric loads and electricity prices exhibited weak-to-intermediate correlations with Pearson correlation indices ranging between 0.30 (Load at SCE, lowest) and 0.39 (SDGE, highest). In contrast, gas prices exhibited relatively stronger correlation with the electricity prices, with Pearson correlation indices ranging between 0.48 (SCE, lowest) and 0.75 (PG&E, highest).

As seen in Table 2, weather information was not provided by the challenge organizers, although being typically incorporated in short-term electricity price forecasting models (Kapoor and Wichitakorn, 2023). Participants were explicitly encouraged to incorporate external data into their forecasting models. The inclusion of exogenous information that was not initially provided in the training set was intended to stimulate the creativity of challenge participants to design innovative and unique modeling solutions and explore various influencing factors in their predictive models, instead of being confined to a specific subset of variables that all teams would be using.

The test dataset follows immediately after the last data point in Figure 2(c), and spans the first two weeks of 2023 (a total of 14 days). It included all information in the training dataset, except for the target variable, namely DA_LMP_PGE_NP15. Despite the fact that the values of the exogenous variables were given for the full 14 days, participants were explicitly asked to only use historical exogenous information up to the time of forecast. In other words, when the model is executed at 8:00 AM each day, the model has to solely use information that is observed up to (and not including) the values of the exogenous variables and electricity prices observed at 8:00 AM. This is to ensure that the models are representative of a realistic day-ahead electricity price forecasting setup.

2.3. Forecasting problem setup and evaluation criterion

In this section, we introduce some notation to formally present the forecasting problem at hand. Specifically, we use a p -dimensional vector, $\mathbf{X}_t = (X_{1t}, \dots, X_{pt})^T$, to denote the values of the exogenous variables (beside the target forecast

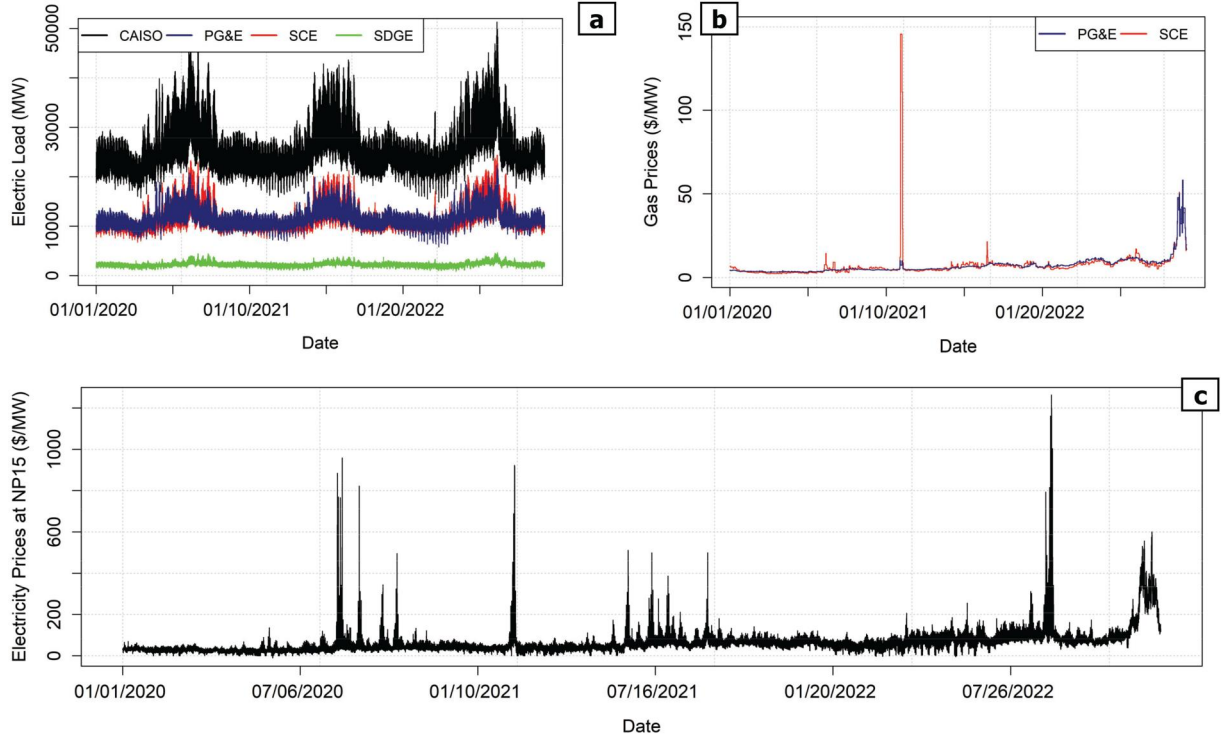


Figure 2. Time series of electric load (Panel a), gas prices (Panel b), and electricity prices at NP15 (Panel c).

variable) at time t such that X_{pt} is the value of the p th exogenous variable at time t . In our training dataset, there are 10 exogenous features, that is, $p = 10$ (refer to the list of variables in Table 2). Additional exogenous variables introduced by the contestants that are not included in the original training dataset are denoted by a q -dimensional vector, $\tilde{\mathbf{X}}_t = (\tilde{X}_{1t}, \dots, \tilde{X}_{qt})^T$. For example $\tilde{X}_{1t}$ could denote the value of the air temperature or grid-scale energy storage at time t . The value of the forecast variable DA_LMP_PGE_NP15 at time t is denoted by y_t . The date and time variables, namely OPR_DATE and HOUR_ENDING, were combined to form the time index $t \in \mathbb{Z}^+$.

Figure 3 presents a high-level schematic of the forecasting problem setup. Teams were tasked with forecasting hourly electricity prices up to 24 hours ahead for each of the 14 days in the test set. Let t_d^* denote the first time index at which forecasts are required for the d th day of the test set, where $d \in \{1, \dots, 14\}$. The vector of forecasts submitted by each team is denoted by $\hat{\mathbf{y}}$, defined as $\hat{\mathbf{y}} = (\hat{y}_{t_1^*}, \dots, \hat{y}_{t_1^*+23}, \hat{y}_{t_2^*}, \dots, \hat{y}_{t_2^*+23}, \dots, \hat{y}_{t_{14}^*}, \dots, \hat{y}_{t_{14}^*+23})^T$, where, for example, $\hat{y}_{t_2^*}, \dots, \hat{y}_{t_2^*+23}$ represent the predicted prices for the 24 hours of day 2. It is important to note that at each day, only one prediction task is carried out, and this single prediction task provides the forecast prices for the following 24 hours. The competition challenged participants to develop a forecasting model, $f(\cdot)$, that uses past pricing data and historical exogenous information to forecast day-ahead electricity prices at NP15. In statistical parlance, the forecasting problem can be broadly expressed as:

$$f: \mathcal{D}^{tr} = \{\mathbf{X}_t, \tilde{\mathbf{X}}_t, y_t\}_{t=1}^{t_d^*-1} \rightarrow \hat{y}_{t_d^*}, \dots, \hat{y}_{t_d^*+23} \quad \forall \quad d \in \{1, \dots, 14\}, \quad (1)$$

where $\mathcal{D}^{tr}$ denote the set of all data used to train the forecasting model $f(\cdot)$. It is also worth noting that the focus of the challenge was primarily on issuing point forecasts—i.e., forecasting intervals or distributions were not explicitly required, although they were indirectly rewarded through evaluation criteria that recognize submission quality and completeness.

Teams were asked to submit their forecasts, along with their documented codes and a 10-page report that detailed their methods. Rankings were determined based on five criteria. The first criterion was forecast accuracy, as measured by Mean Absolute Error (MAE), which constituted 40% of the final score. Although forecast accuracy was explicitly mentioned as a primary evaluation criterion when announcing the competition, the specific evaluation metric to assess forecast accuracy was not disclosed to the contestants. This was primarily intended to incentivize teams to design forecasting models that generalize well in terms of average deviation, but that are also unbiased and robust to pricing spikes.

The second and third criteria, each comprising 20% of the final score, were simplicity and novelty, respectively. These criteria encouraged teams to infuse creativity in their model development, while avoiding unnecessary complexity. The fourth criterion was the inclusion of additional features, representing 10% of the final score. This criterion was especially introduced to encourage teams to search and introduce new variables beyond those provided by PG&E. The final criterion is methodology description, representing another 10% of the final score, which encouraged participants to clearly articulate the rationale behind their chosen methodology and why it is well-suited for the problem at hand. This is primarily to ensure that the models developed

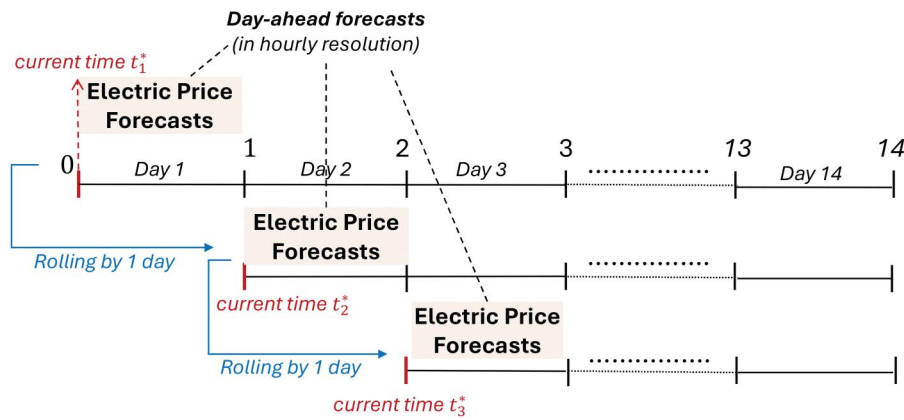


Figure 3. Problem setup for the 2024 IISE PG&E Energy Analytics Challenge.

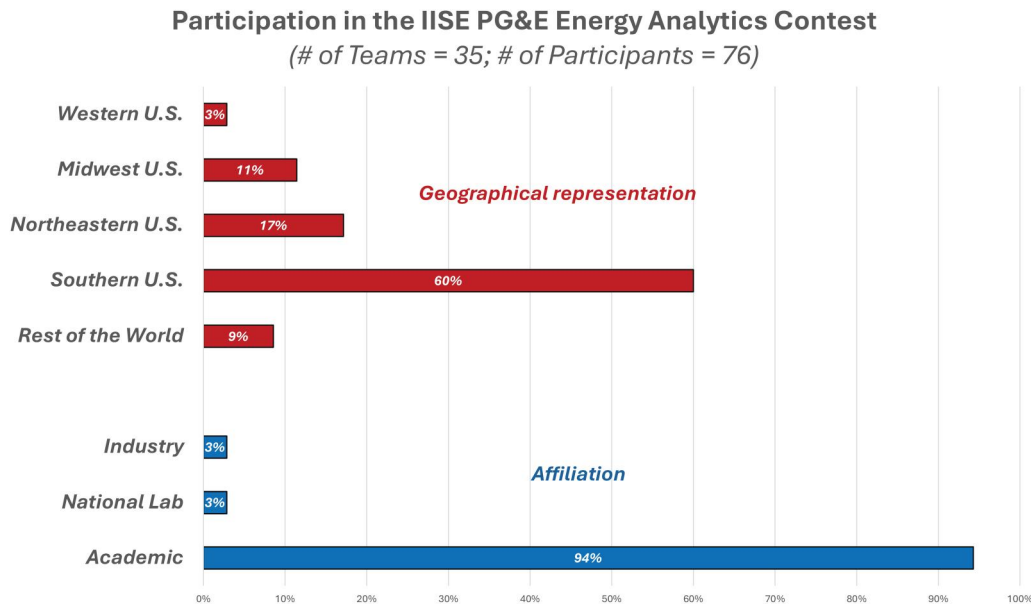


Figure 4. Geographical representation and affiliation of the participating teams in the 2024 IISE PG&E Energy Analytics challenge.

by the participating team can benefit future research and development at PG&E. Collectively, evaluation criteria aimed to not only reward predictive accuracy but also solution ingenuity, simplicity, and a strategic selection of variables that contribute to the overall effectiveness of the proposed forecasting model.

Except for the accuracy scores (first criterion), blind evaluations were carried out by a team of six industry-academia experts. Three of the evaluation committee members were PG&E staff who are involved in the electricity market business. The evaluators reviewed all submitted materials independently and provided their scores. The challenge organizers then aggregated all scores and ranked the teams from highest to lowest.

3. Summary of methods and results

A total of 35 teams participated in the competition, comprising 76 participants. Each team had an average of 2.17 members. Figure 4 presents the distribution of the teams, in terms of geographical location and affiliation (e.g., academia, U.S. national lab, or private sector). The large majority of

teams were based in the U.S., with a fairly diverse representation between the west, east, northeastern, mid-west, and southern U.S. About three teams (out of 35) were from non U.S.-based institutions. The overwhelming majority of the contestants were academic teams, either formed of students (typically doctoral students at academic research institutions), or faculty-student teams representing a research group at an academic institution.

The three top-ranked teams were invited to present their findings at a special session at the IISE Annual Conference, which was held in May 2024 in Montreal, Canada. Below, we briefly present a high-level description of the methods submitted by those top-performing teams. We only present the high-level details in order to not compromise the authors' ability to publish or possibly commercialize their solutions, if they so desired.

3.1. Team 1: An LSTM Approach with Exogenous Information

This team was based at the University of Miami's Data Analytics Laboratory and included four members: Salih

Salihoglu, Afshin Asadi, Ibrahim Ahmed, and Ramin Moghaddas. The proposed approach was mainly based on a Long Short-Term Memory (LSTM) model with several exogenous features that are tailored to electricity price forecasting. An LSTM is a special architecture of DNNs that is suited for modeling time series data. The authors used a validation set comprising the last 3 months of the training dataset to fine-tune the architecture parameters (including the number of layers, neurons, and regularization techniques) and assess its generalization capabilities.

A key strength of this team's approach is the inclusion and handling of a large set of exogenous covariates beyond those supplied by the challenge organizers. In addition to the load and gas price information provided in the original training dataset, the following list of additional features were used as inputs to their proposed LSTM-based model:

- *Weather Data*: Information and forecasts about spatio-temporal meteorological conditions across California, such as temperature, precipitation, humidity, and wind speed.
- *Grid Status Data*: Information and forecasts about the variability in the energy generation mix over time, including renewable sources, conventional power generation, and energy storage.
- *Temporal Features*: Time-based features, such as sinusoidal transformations of hours, days of the week, and months, alongside indicators for weekends and specific days, were created and added to the model to capture the cyclical nature of electricity demand and pricing.

The team further improved their model by introducing a novel custom loss function to enhance the prediction accuracy and interpretability. An online learning approach has been developed to allow the model to adapt to new data incrementally, ensuring continuous relevance and accuracy. The inclusion of those customized features and exogenous covariates turned out to be a major contributor to improving the performance of their LSTM model. In addition to time series residual diagnostics, the authors reported an MAE of 13.94 (\$/MW) and an RMSE of 23.10 (\$/MW) on their validation set. Their final MAE and RMSE values, evaluated by the challenge organizers on the blind test set, were found to be 18.92 (\$/MW) and 24.39 (\$/MW), respectively. Those values are fairly close to those reported by the authors, which indicates that their train-test experiments were well-executed.

3.2. Team 2: A multi-stage tree-based ensemble model

This team was based at Syracuse University and included three members: Shivakumar Suresh, Rahul Makarand Kulhalli, and Natarajan Gautam. The team implemented a multi-stage tree-based ensemble approach. In the first stage, a Random Forest (RF) regression model was trained directly on the historical pricing signals and exogenous information to generate day-ahead forecasts. The residuals of the model were then extracted. A gradient boosting model was then fit

directly to the residuals of the RF model. The final prediction is computed as the sum of the predictions from the two additive model components.

This team also included exogenous weather and time-based features. Their variable importance analysis concluded the integration of the following features: forecasts of rain, temperature, and holiday information. Aligning with the conclusions of Team 1, weather information was found to have strong explanatory power in describing the variations in electric pricing. However, no external grid or operational information was included by this team, beyond those provided in the original dataset. The team performed several validation experiments by comparing their ensemble approach against baseline methods such as autoregressive-based models, linear regression, LSTMs, transformers, among others. One important observation made by this team is that increasing model complexity often leads to severe overfits, especially with transformer-based models. The team reported an MAE of 4.81 (\$/MW) on their chosen validation set. Their final MAE value evaluated by the challenge organizers on the blind test set, was found to be 29.08 (\$/MW). Despite a noticeable divergence between their reported and actual performance, Team 2 fared well in terms of forecast accuracy, and ranked second in terms of overall rank due to the innovation and simplicity of their approach, which was well-commended by the evaluation committee.

3.3. Team 3: A statistical machine learning approach: Fusing vector autoregressive methods with tree-based ensembles

This team was based at Oklahoma State University and included two members: Enrico Laoh and Boris Oskolkov. This team relied on a hybrid of a traditional statistical learning approach with an ML technique. The motivation behind this hybrid approach was to borrow strength across the two distinctive paradigms. Specifically, the team started by a Vector AutoRegressive (VAR) approach to model the multivariate time series provided in the training dataset jointly. This model was further bolstered by integrating advanced ML techniques, including Gradient Boosting Tree, and Extreme Gradient Boosting (XGBoost), with XGBoost emerging as the most effective in terms of forecasting accuracy.

After observing intermediate-to-high correlations between gas prices and electricity prices, the team included an additional external variable, which describes natural gas pricing in California. Unlike the first two teams, which considered several external factors, this team relied solely on this exogenous feature and did not incorporate any exogenous weather information. By feeding the ML models with variables transformed through VAR, the team was able to demonstrate the improved predictive accuracy of their model, with a reported MAE of 2.74 on their test set. Their final MAE value evaluated by the challenge organizers on the blind test set, was found to be 26.82 (\$/MW).

3.4. Comparative analysis and benchmarking

Figure 5 shows the time series of the forecasts from the top three teams. The forecasts are closely aligned with the actual data that was withheld from the contestants. Visually, Team 1's forecasts appear to be the closest to the actual data, which aligns with the quantitative evaluation comparisons. Teams 2 and 3 seem to perform similarly, with the former slightly over-predicting at times, whereas the latter, in contrast, exhibits some under-prediction bias. In addition, we compare the performance of the proposed methods against four representative benchmarks in the time series literature, which are listed below:

- Naïve forecast: This model assumes that the most recently observed electricity price will persist into the future. Despite its simplicity, it is a standard benchmark in the forecasting literature and tends to perform well for ultra-short-term horizons. It is expressed as:

$$\hat{y}_{t_d+h} = y_{t_d-1} \quad \forall \quad h \in \{0, \dots, 23\}. \quad (2)$$

- Seasonal naïve forecast: Unlike the naïve model, the seasonal naïve approach assumes that the value observed during the most recent season will persist in future seasons. It generalizes the naïve model for modeling seasonal data, and is expressed as:

$$\hat{y}_{t_d+h} = y_{t_d+h-m(k+1)} \quad \forall \quad h \in \{0, \dots, 23\}, \quad (3)$$

where m is the seasonal period, and k is the integer part of $\frac{h-1}{m}$. In our context, we let $m = 24$ hours, since the electric pricing data exhibits diurnal trends. In the electricity price literature, this is often referred to as the *same-day* approach (Weron, 2014).

- Autoregressive Integrated Moving Average Model (ARIMA): The ARIMA model is one of the widely used time series approaches where the (differenced) time series of the target variable is regressed on lagged values of the predictions as well as lagged values of the error. It is expressed as:

$$\hat{y}'_{t_d+h} = \phi_0 + \sum_{i=1}^p \phi_i \hat{y}'_{t_d+h-i} + \sum_{\ell=1}^q \theta_{\ell} \epsilon_{t_d+h-\ell} + \epsilon_{t_d+h} \quad (4)$$

$$\forall \quad h \in \{0, \dots, 23\},$$

where $\{\phi_i\}_{i=0}^p$ and $\{\theta_{\ell}\}_{\ell=1}^q$ are model coefficients, and $\epsilon \sim \mathcal{N}(0, \sigma^2)$. The two parameters p and q are the orders of the autoregressive and moving average terms, respectively, whereas y'_t is the differenced time series at time t . The differencing order, r , is another model parameter. All parameters are estimated using the `auto.arima()` package in R, which combines stepwise selection, Akaike Information Criterion (AIC), and Maximum Likelihood Estimation (MLE) to determine the best-fitting ARIMA model (Hyndman and Khandakar, 2008).

- Autoregressive Integrated Moving Average Model with Exogenous Features (ARIMA-X): The ARIMA-X model is a generalization of the ARIMA model that incorporates exogenous features as linear regressors. It is expressed as:

$$\hat{y}'_{t_d+h} = \phi_0 + \sum_{i=1}^p \phi_i \hat{y}'_{t_d+h-i} + \sum_{\ell=1}^q \theta_{\ell} \epsilon_{t_d+h-\ell} + \epsilon_{t_d+h} + \sum_{j=1}^p \beta_j \hat{X}_{t_d+h-\ell} \quad \forall h \in \{0, \dots, 23\}, \quad (5)$$

where $\{\beta_j\}_{j=1}^p$ are regression coefficients, and $\hat{X}_{t_d+h-\ell}$ are 24-hour ahead predictions of the exogenous features, which are assumed to be predicted using a naïve model. The parameters were also estimated using the `auto.arima()` package in R. Both ARIMA and ARIMA-X were trained using all the provided data.

The baseline models were chosen to represent classic, yet widely prevalent, forecasting models in the literature and practice. The implementation of all benchmark models is straightforward and we closely followed the standard implementation of those models as detailed in Hyndman and Athanasopoulos (2021). To ensure fair comparison, all models were trained and tested on the same datasets. Table 3 shows key performance metrics, namely MAE, RMSE, and forecast bias (computed as the signed difference between actual and predicted values), for all the proposed models from Teams 1 through 3, as well as those from the representative benchmarks presented above. We see a clear superiority of the proposed models from the three teams relative to

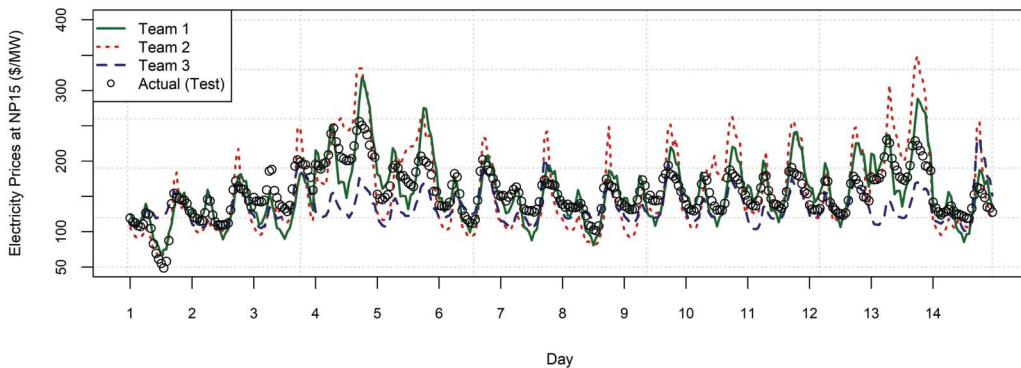


Figure 5. Time series of 14-day forecasts from top-performing teams relative to the actual test data which were masked from all participants.

Table 3. Final scores and rankings of competing teams. Bold-faced values indicate best performance

Benchmark	MAE	RMSE	MAPE%	Bias
Team 1 (LSTM with exogenous features)	18.92	24.39	12.04	−1.85
Team 2 (Tree-based ensemble)	29.08	36.03	18.16	−2.08
Team 3 (VAR + XGBoost)	26.82	35.17	17.08	20.2
Naïve	41.01	50.36	24.70	37.9
Seasonal Naïve (period = 24 hours)	38.0	46.3	23.3	35.3
ARIMA (no exogenous features)	38.4	48.1	23.1	35.1
ARIMA-X (w/ exogenous features)	32.3	44.0	21.5	4.5

the baseline models across almost all benchmarks. We provide a more in-depth discussion in Section 4 about our observations regarding where and how did the evaluation committee perceive from where did those improvements stem.

4. Discussion and lessons learned

In this Section, the key insights and lessons learned are summarized, based on the analysis of the results presented in Section 3. Those are listed in the following itemized summary, along with their relevance to the topic of electric price forecasting:

- **Tailoring ML/AI models is the key to unlocking the promise of emerging predictive methods:** It was found that ML/AI models improve significantly over traditional time series methods which are often limited by restricting distributional and parametric assumptions. On average, forecasts from the three teams improved over those from a simple naïve model by 39.19% (in terms of MAE) and by 22.79% relative to an ARIMA-X that uses all exogenous features provided. However, the caveat here is that off-the-shelf ML/AI methods often do not perform well if not carefully tuned and tailored to the application of interest. In contrast, tailored and contextualized ML/AI approaches that include customized aspects such as context-specific exogenous information, careful and domain-informed feature engineering and selection, custom loss functions, and potentially a potent combination of more than one ML/AI model, is the key to unlock the value of emerging predictive techniques for short-term electricity price forecasting. Extensive validation to fine-tune ML/AI parameters was also noted across all teams. In particular, Team 2 thoroughly investigated this in their report, where they compared their customized multi-stage tree-based ensemble method to various off-the-shelf LSTMs and transformers. They demonstrated that, despite being more sophisticated, these models can significantly overfit and underperform due to their inability to distinguish between systematic signals and unexplained variations in the historical pricing data. In summary, ML/AI model, have immense potential, but require significant effort to contextualize and optimize their performance.
- **Including exogenous information in an electricity price forecasting model is crucial:** The results in Table 3 show that including exogenous features is extremely

Table 4. The value of exogenous information in electric price forecasting

Team	# of added features	Weather?	Grid?	Temporal?	Feature choice	Skill%
1	46	✓	✓	✓	✓	53.9
2	4	✓	–	✓	✓	29.1
3	1	–	✓	–	✓	34.6

important to the effectiveness of electricity price forecasting models. All the top three teams included some external information that was not provided in the raw PG&E dataset. Even ARIMA-X, despite falling behind the proposed models from the three teams, performed significantly better than its counterpart without exogenous features (15.88% improvement in terms of MAE). The use of a large and informative exogenous feature set appears to have been Team 1's trump card. Table 4 shows the comparison of the three top-performing teams in terms of the number and type of features that were added beyond those that were originally provided in the raw dataset. We also compute the skill score of all teams, defined as in (6).

$$Skill = 100 - 100 \times \frac{MAE(\text{proposed})}{MAE(\text{naïve})}. \quad (6)$$

We notice that Team 1 has a significantly larger number of features included (total of 46 added features) relative to Teams 2 and 3 which only had four and one added feature(s), respectively. This inclusion of a significantly larger feature set has been highly rewarding to Team 1's performance, as evident by the significant increase in skill score. Added features ranged between weather-related information (e.g., temperature, wind conditions, rain), grid condition data (e.g., generation mix, available storage, imports), and time-based features (e.g., holiday information, day of the week, and weekday versus weekend). All teams had some element of feature selection, either as an independent step prior to model training, or through the model's performance on an independent validation set. This finding aligns with the consensus in the recent electricity price forecasting literature, which suggests that introducing larger feature sets is almost always beneficial, so long that properly designed feature selection techniques are in place to balance model complexity and generalization capability (Wang *et al.*, 2017; Kapoor and Wichitaksorn, 2023).

- **Hybrid and ensemble models have significant potential:** We observe that two of the top-performing teams used some form of a hybrid or ensemble method. Team 2 employed a sequential hybrid approach using two prevalent ML techniques, namely RFs and XGBoost, both of which are ensemble learning methods. Thus, their approach can be seen as an ensemble of ensembles. Team 3, on the other hand, hybridized a classic multivariate time series method (namely, VAR) with a modern ML approach, XGBoost. This underlines the value of fusing contrasting paradigms of time series modeling, and how statistical and ML/AI paradigms can complement, rather than compete against, each other. In fact, the

observation of combining multiple data-driven models have been noted in several energy-focused forecasting problems, for example by combining copulas with support vector machines (Arrieta-Prieto and Schell, 2022), transformers with kernel-based models (Ye *et al.*, 2024b), or through sums of independent Gaussian Processes (Prakash *et al.*, 2023), and combinations of time series models (Nowotarski *et al.*, 2014). It is also both refreshing and reassuring to observe that the importance of model hybridization and ensemble learning is aligned with the findings of the M competitions, even beyond electricity price forecasting, which have also suggested that forecasting accuracy can be significantly enhanced by combining distinct modeling techniques (Makridakis *et al.*, 2022b).

5. Conclusions and recommendations

In this article, we presented the 2024 IISE PG&E Energy Analytics challenge organized by two IISE divisions in collaboration with a major utility in the U.S., PG&E. The challenge focused on forecasting day-ahead electricity prices using real-world pricing signals and exogenous information from California's electricity market. A total of 35 teams participated, comprising 76 members. The top three teams were identified, primarily based on the accuracy of their forecasting models, but also through other criteria such as innovation, creativity, and simplicity of their proposed solutions. The challenge serves as an exemplar of academic-industry collaboration towards promoting the science and practice of energy forecasting through blind and fair benchmarking and testing of various data-driven solutions. Despite the value of our post-challenge analysis presented in this article, we acknowledge that extensive testing on longer test sets may be needed to fully confirm our findings. We therefore hope that future offerings of this competition (or other similarly focused contests) would identify points of parity and difference and investigate whether similar conclusions are reached, across different datasets, models, and environments.

We summarize the main findings and lessons learned drawn through key common themes that we have deduced across the top-performing teams. The customization and contextualization of ML models, including additional exogenous information, and the hybridization of several techniques are the most important insights drawn. Reflecting on the 2024 challenge, few recommendations for other similar competitions in the future come to mind:

1. *Rolling/Multi-round competitions:* Due to the initial time investment of introducing the inaugural version of this competition, the challenge organizers decided to have only one round of submission for this competition in order to provide ample time for the teams to design and test innovative ideas. However, we do acknowledge that multi-round competitions might be more suited for short-term forecasting problems, where newly revealed information is provided to the contestants at every new submission milestone. Contestants can use the newly revealed information to calibrate and modify their models, whereas the final scores can be computed as a (weighted) aggregate of all rounds, while ensuring effective procedures to avoid data leakage.
2. *More international dissemination:* Although the competition was intended to attract international participation, an overwhelming majority of the participation came from U.S.-based institutions. Dissemination beyond the U.S. requires an extensive leveraging of non-traditional platforms beyond mailing lists and professional society's websites such as social media, as well as personalized e-mails to connections at non-U.S. universities. Non-U.S. representation in the organizing committee can also help spread the word about the registration for the competition. It is worth noting that a more diverse representation, while definitely desirable, may raise issues of accessibility to computational resources, which could, in turn, have an influence on model design, method choice, as well as the ability of teams from underprivileged institutions to unlock access to more complex and powerful models. One potential solution is to offer cloud-based GPU compute credits by the challenge sponsors that would be available to teams, at no charge, upon request.
3. *Collaborations with other disciplines:* Forecasting of electricity prices is a multi-faceted problem with contributions from the broader engineering, business, and data science research communities. One way to increase the participation and diversity of the challenge contestants is through joint organization and offering of such competitions, with other professional divisions and societies within, for example, the Institute of Electronic and Electrical Engineers (IEEE), the International Institute of Forecasting (IIF), and the Institute for Operations Research & the Management Sciences (INFORMS).
4. *Evaluation:* The results in Table 3 suggest that some methods may experience vastly different performance, depending on which error measure is used. Future competitions will look into standardizing the use of error metrics in order to provide a more holistic evaluation of a method's capability. Moreover, the evaluation metric would be chosen in light of the context of the application. For example, utilities (like PG&E) as well as other electricity market stakeholders are often interested in accurately predicting "extremes," due to their serious economic implications. This includes spikes in electricity prices (as those shown in Figure 2(c), peak load (Shukla and Hong, 2024), or wind energy ramps (Ezzat *et al.*, 2019). In those contexts, unconventional error metrics such as peak shape error (Shukla *et al.*, 2024), weighted evaluation metrics, or even classification-based measures (Sheybanivaziri *et al.*, 2024), may be invoked. Another direction pertains to the increasing interest in probabilistic forecasting in both the literature and practice of energy systems (Nowotarski and Weron, 2018; Sweeney *et al.*, 2020), which calls for introducing requirements on reporting probabilistic information

such as prediction quantiles, as well as the use of scoring metrics to evaluate the quality of the ensuing probabilistic predictions, for example, using Pinball loss or continuous ranked probability scores. Finally, it is worth noting that four out of five evaluation criteria in this challenge were based on qualitative aspects such as methodology description, novelty, and simplicity. While we believe in the importance of these criteria, future offerings of this competition will ensure a more precise description of what each criterion entails towards a more objective assessment of all submissions.

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Data availability statement

The data used for this challenge, along with a brief description, are publicly available on GitHub at <https://github.com/RIA-Research-Group/2024-IISE-PGE-Electricity-Price-Forecasting-Challenge.git>.

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